

Challenges in Teaching Process Mining: Insights from Process Mining Educators

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Abstract. Process mining has gained traction in both academia and industry, enabling organizations to discover, monitor, and improve their work processes using event logs generated by information systems. As demand for process mining grows, so does the need for skilled professionals who can apply it and interpret the results effectively. While process mining educational programs are evolving, ranging from university courses to industry training programs, little is known about the specific challenges educators face in teaching process mining. This study aims to close this gap by investigating key challenges encountered by process mining educators. Based on a series of focus groups involving experienced process mining educators from both academic and industrial settings, we identify nine challenges spanning diverse teaching tasks. These findings provide valuable insights for developing more effective teaching strategies and resources to support process mining education.

Keywords: Process Mining · Education · Teaching Challenges.

1 Introduction

Process mining has emerged as a rapidly growing field, enabling organizations to discover, monitor, and improve work processes using event logs generated by information systems and applications [12]. As process mining adoption continues to gain traction across various industries, the demand for skilled process mining professionals rises accordingly. However, developing process mining expertise is not trivial, as it requires a broad set of competencies, spanning technical, analytical, and business-oriented skills [9].

Given the interdisciplinary nature and the diverse competencies required, process mining educators must equip learners not only with tool-specific knowledge and algorithmic foundations but also with the ability to interpret and apply process mining outcomes within real-world organizational contexts. Recent studies identified the lack of domain expertise, analytical reasoning, and technical proficiency as some of the most pressing challenges associated with the successful implementation of process mining initiatives [10,14]. Furthermore, one of

the success factors in process mining is a proper training with balance of both technical and business dimensions. A diverse skillset is expected from practitioners, which spans programming, data analysis, project management, and business process understanding [6].

Despite the growing availability of process mining education through university courses, commercial training programs, vendor-specific certifications, or internal company workshops, little is known about the specific challenges encountered by process mining educators.

In this work, we address this gap by empirically investigating the key challenges in teaching process mining. To do so, we conducted a qualitative study based on focus groups with process mining educators from both academia and industry. This method allowed us to capture varied perspectives on challenges across different educational settings. We organize these challenges according to key teaching tasks, from the identification of learners' needs to implementation and evaluation of courses to inform future work aimed at supporting educators in enhancing process mining teaching tasks. To the best of our knowledge, this is the first empirical study that investigates educational challenges associated with teaching process mining.

This paper is structured as follows. Section 2 discusses the background and research gap. Next, Section 3 details our research method, followed by Section 4, which presents our findings. In Section 5, we provide research directions based on our main findings. Finally, we conclude the paper and outline future work as well as limitations of our study in Section 6.

2 Background

In this section, we introduce the background and research gap. First, we outline relevant aspects about teaching process mining. Then, we provide an overview of how process mining has been used in education.

2.1 Teaching Process Mining

Process mining as a discipline is offered through a wide range of formats. In academic settings, it is typically integrated into courses related to information systems, business process management, or data analytics, where it may be taught as part of a broader curriculum or as a dedicated module³. Depending on the disciplinary focus of the program, technical and analytical or managerial aspects can be emphasized more. In technical programs, such as in computer science or data analytics, the focus is often on the underlying algorithms, data processing techniques, and the use of specialized process mining tools. On the other hand, in managerial or business programs, process mining is typically framed within the context of process improvement, operational efficiency, and strategic decision-making. This flexibility allows process mining to be adapted to diverse academic disciplines.

³ <https://www.tf-pm.org/resources/education>

In industry contexts, training often takes different forms. For instance, there are tool-focused courses offered by process mining vendors such as Celonis ⁴, Fluxicon ⁵, among others. In addition, self-paced online courses are available, ranging from general introductions to more advanced topics ⁶. Beyond these semi-public offerings, many companies also develop their own internal training programs and materials as part of process mining adoption efforts, often tailored to specific business processes, systems, and organizational goals.

In this study, we use a broad definition of the term *educator*, referring to all individuals involved in teaching and facilitating the learning of process mining concepts, skills, and tools. This includes educators who are involved in teaching activities mentioned above, university lecturers and professors teaching formal academic courses, as well as corporate trainers, consultants, and practitioners responsible for professional development sessions, workshops, or internal training programs. Throughout this study, we use the general term *educational course* to refer broadly to any structured learning offering. This includes traditional academic courses, workshops, professional training sessions, and other educational products designed by process mining educators to teach process mining.

The diversity of educational formats and educators highlights the interdisciplinary and practice-oriented nature of the process mining field. As process mining continues to evolve and gain traction across both academic and industry contexts, gaining insight into the challenges involved in teaching process mining becomes increasingly important. Such understanding is essential for developing effective educational strategies, aligning learning objectives with stakeholder needs, and supporting the broader adoption and meaningful application of process mining in various domains.

With this paper, our goal is to address this gap by identifying and understanding the key challenges faced while teaching process mining.

2.2 Process Mining in Education

Academic research has predominantly studied process mining in education through an application-driven lens, focusing on how to use process mining techniques to analyze educational processes. Researchers have explored process mining use in analyzing student behavior, enhancing learning processes, and improving curriculum design [2]. For example, Wambsganss et al. [16] propose a framework and taxonomy for leveraging process mining in learning environments to guide the discovery, monitoring, and enhancement of student learning processes to improve the individualization of education at scale. Several studies focus on how to apply process discovery or conformance checking techniques specifically to analyze the learners data [1,2,4,11]. However, all mentioned studies focus on process mining in education and not on the challenges involved in teaching others about process mining, which is the focus of our work.

⁴ <https://academy.celonis.com/courses/symbio-kickstarter>

⁵ <https://fluxicon.com/camp/>

⁶ <https://www.coursera.org/learn/process-mining>

To the best of our knowledge, only one study addresses challenges in teaching process mining by exploring gamification to enhance motivation and engagement. Delcoucq proposes using game elements [5] to improve student participation and understanding. Similar approaches have been explored in data science and statistics education to tackle issues like abstract content, diverse learner backgrounds, and limited engagement [8]. Broader challenges, such as aligning theory with practice and sourcing suitable datasets, are common in teaching data science [7]. While broader challenges in data science education and gamification are relevant, our study focuses on identifying challenges specific to teaching process mining.

3 Research Method

To explore the challenges involved in teaching process mining, focus groups were selected as the primary data collection method [3]. Given that the participants come from both academia and industry, the focus group setting allowed us to observe how their perspectives align or diverge on specific challenges. The interactive nature of focus groups enabled participants to build on each other’s responses, leading to deeper insights into shared concerns as well as contrasting experiences. Figure 1 provides an overview of our 4-step study design. We facilitated three focus groups. After each focus group, we conducted data analysis sessions followed by a validation session with selected participants to verify our findings and gather further feedback, ensuring reliability of our results. Next, we describe the definition of the target population, the procedure we followed to collect the data, and how we analyzed and validated the data.

3.1 Definition of Target Population

Since our aim was to gain a comprehensive understanding of experiences in teaching process mining, we purposefully recruited participants from both academia and industry. To do so, we applied three alternative inclusion criteria to ensure participants had teaching responsibilities, even if in different settings. Specifically, eligible participants included: (1) academics teaching process mining in

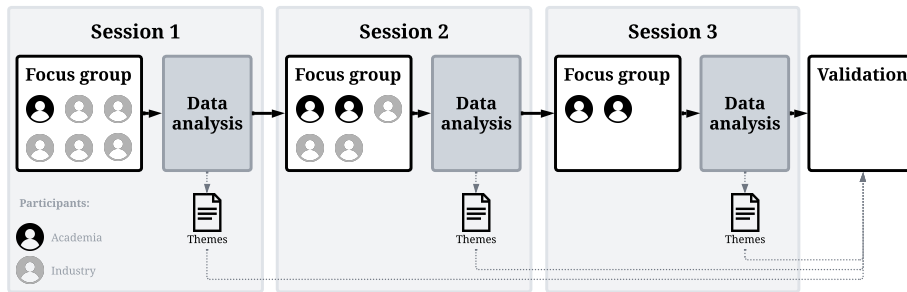


Fig. 1. Overview of our study design.

higher education; (2) industry practitioners, such as consultants, delivering training or enablement sessions for clients; and (3) industry practitioners teaching process mining internally, for example, as part of onboarding or internal knowledge transfer programs. We invited participants from our professional network and sent invitations via LinkedIn or email.

Table 1 shows the participants demographics, including focus group allocation and years of process mining experience. All participants conduct their teaching activities mainly in Europe.

3.2 Data Collection

We facilitated the focus groups and a validation session in an online manner using MS Teams as our video conferencing tool. Each focus group lasted approximately 90 minutes, and the validation session took 45 minutes. In total, we had 13 participants in the focus groups: eight from industry and five from academia. Out of the eight participants from industry, one also has a PhD in process mining, and has had experience in teaching process mining in academia.

Each focus group session started by informing participants about the study’s research objectives and goals. This was followed by a brief introduction round, during which participants were asked to share their current role, the number of years of experience they had with process mining, and the specific teaching activities they had undertaken. Following the introductions, participants were asked to individually reflect on their personal teaching experiences and write down the specific challenges they had encountered. To gain insight into the per-

Table 1. Participants’ demographics, with “FG” column indicating the focus group assignment, and “Exp.” column indicating years of process mining experience.

FG	P	Area	Current Role	Exp.
1	1	Academia	Assistant Professor	11-15
	2	Industry	Process Mining Consultant	1-5
	3	Industry	Process Intelligence Consultant	1-5
	4	Industry	Process Intelligence Consultant	6-10
	5	Industry	Consultant	11-15
	6	Industry	Director of Process Mining Success	16-20
2	7	Academia	Assistant Professor	6-10
	8	Academia	Associate Professor	11-15
	9	Industry	Data Science Lead	1-5
	10	Industry	Senior Process Mining Consultant	6-10
	11	Industry	Global Process Excellence Owner	6-10
3	12	Academia	PhD Candidate	6-10
	13	Academia	Assistant Professor	11-15

ceived difficulty of tackling each challenge, we also asked participants to rate their challenges using a three-point scale to indicate whether the challenge was *somewhat challenging*, *mostly challenging*, and *extremely challenging*. After the individual reflection, we facilitated a group discussion where participants shared and elaborated on their challenges perceived as *extremely challenging*. This approach allowed for grounding the group discussion in individual experiences, offering an initial sense of which challenges were most pressing across participants. The collaborative dialogue allowed for deeper exploration of common issues and contrasting perspectives.

Inspired by [15], we followed the principle of saturation to determine when additional focus groups were no longer yielding new or meaningful insights.

3.3 Data Analysis and Validation.

Following each focus group, we transcribed the video recording and used qualitative thematic coding [3,13] to identify recurring patterns and unique insights. All individual challenges were aggregated into a consolidated dataset. After that, the coding was performed iteratively by the first and last authors. Both began with independent “in-vivo” coding to stay close to participants’ original used language. Through multiple rounds of discussion and refinement, the authors developed a shared coding scheme, grouped similar codes (e.g., *Difficulty to obtain good and diverse dataset for training* and *There is some data available, but not specifically for training*) into broader categories (e.g., *Data availability for training*), and identified overarching themes (e.g., *Finding or generating suitable data and event logs*). Divergences in coding were resolved through collaborative discussion until consensus was reached.

After analyzing the acquired data from the second focus group and comparing it with the first, we observed a clear decline in the new challenges, with no additional high-level themes emerging, indicating we were likely approaching theoretical saturation. To confirm this, we conducted a third focus group, which reinforced our earlier findings, yielding no new high-level challenges. After the third and last focus group, we conducted a validation session with selected participants to review and confirm the relevance and completeness of the identified themes. Across all sessions, participants shared 92 individual challenges, which resulted in nine high-level themes, presented in detail next.

4 Findings

In this section, we present the findings of our study. Figure 2 provides an overview of the high-level challenges identified, along with the number of participant-reported individual challenges that support each of them. For reporting purposes of our findings, we structured the challenges into three high-level categories related to teaching tasks: (i) identifying learners’ needs, (ii) design and development of course materials, and (iii) implementation and evaluation of courses. Anyhow, have in mind that some challenges may span multiple categories.

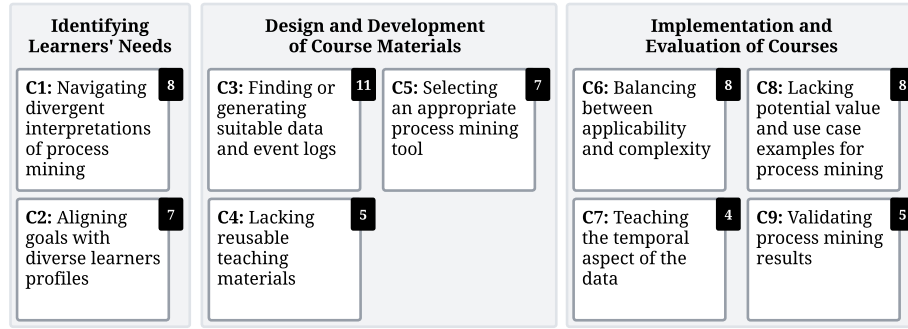


Fig. 2. Challenges in teaching process mining.

4.1 Identifying Learners' Needs

In this phase, educators identify learners' needs and clarify goals, understanding the gap between what learners currently know and what they need to know. We identified two high-level challenges related to this phase.

C1: Navigating divergent interpretations of process mining. We observed notable variability in how participants conceptualize process mining. Participants in our study referred to process mining using different framings, some characterized it predominantly as a tool, while others described it as a set of techniques or a methodological framework. These differing perspectives may have implications for how process mining is taught, particularly during the early stages of curriculum development. Specifically, they influence decisions in teaching materials design, particularly whether to emphasize practical tool use or to focus on the theoretical foundations and algorithmic part of the discipline.

This divergence in process mining conceptualizations is often specific for the distinction between academia and industry, where industry participants tend to adopt a tooling-oriented perspective and academic participants emphasize methodological foundations. However, this distinction is not absolute and some participants from industry critically reflected on the limitations of teaching PM primarily as a tool. For instance, P10 mentioned that an overly tool-centric approach, while initially seems more engaging at the outset due to its practical nature, may limit learners' deeper understanding and ability to use PM for broader, real-world contexts. As P10 emphasized, if learners are taught only how to replicate predefined steps within a tool, for example by mimicking a specific use case, they may become proficient in solving similar tasks, but lack the conceptual understanding necessary to identify new opportunities for PM application or adapt the method to other, unstructured problems.

C2: Aligning goals with diverse learners profiles. One of the challenges reported by both participants from academia and industry is adapting the teaching to different background and skills. Process mining students in academia can come from varied backgrounds, such as computer science or business; each with

distinct strengths and knowledge gaps. For instance, P8 observed that computer science students often excel in data manipulation tasks but may lack a deeper understanding of business process modeling and analysis. In contrast, students who have previously completed courses in business process management tend to have stronger conceptual knowledge of processes, which can help them better integrate both technical and business process perspectives when learning about process mining. Thus, it is challenging for process mining educators to align teaching goals with such a diverse realm of process mining learners' profiles.

This challenge is equally prevalent in industry, where process mining learners come from both technical roles (e.g., data engineers and IT specialists), and business-oriented roles (e.g., process owners and compliance managers). These groups often approach process mining with differing expectations and learning goals. Technical professionals may focus on aspects such as data pipelines, system integration, and tool customization, while business-oriented learners are typically more interested in deriving actionable insights and identifying potential process improvements. Consequently, designing educational course that is relevant, engaging, and appropriately challenging for such a diverse audience remains a significant obstacle in effectively teaching process mining.

4.2 Design and Development of Course Materials

Once the learners' needs and goals are defined, educators move on to educational course materials design and development. This involves selecting or developing appropriate datasets, choosing process mining tools or platforms, and creating supporting materials such as slides and exercises.

C3: Finding or generating suitable data and event logs. Another challenge reported by both industry and academia participants relates to the effort required to find or generate suitable training data for process mining. Educators often need to invest substantial time in creating datasets that are both realistic and appropriate in terms of complexity. Participants emphasized that it is particularly challenging to source data that includes meaningful quality issues, i.e., issues that students can learn to detect and resolve. In many cases, educators must simulate such imperfections manually, which significantly increases the content preparation workload.

We also observed some differences in how this challenge is experienced in academic versus industry settings. In academia, process mining is typically taught using curated event logs designed for teaching purposes, allowing students to focus on process discovery tasks. By contrast, industry practitioners often work with raw data extracted from source systems, typically distributed across multiple relational tables. Consequently, training materials for industry learners must often replicate this particular setting by including not a single event log, but a collection of raw data tables that require transformation into an event log format. Simulating this kind of realistic data scenario for training purposes is even more time-consuming and technically demanding.

C4: Lacking reusable teaching materials. Participants from both academia and industry also reported on a limited availability of reusable teaching materials. Beyond the challenge of finding or generating suitable data and event logs (cf., C3), participants noted a lack of available teaching resources that include practical exercises, guided examples, and case studies. P1 mentioned specifically the lack of interactive training tools or sandbox environments where learners can interactively experiment with both theoretical concepts and practical tasks, such as applying filters, modifying event logs, or selecting algorithms, and immediately observe the consequences of their actions. On the theoretical side, students have few opportunities to explore how different algorithms behave under varying input conditions or assumptions. On the practical side, they may not easily see how incorrect assumptions (e.g., about case identifiers) or improper manipulations (e.g., applying misleading filters) can distort outcomes. As P7 shared, learners need environments where “*they are allowed to make mistakes*”, in a way that helps them understand why something failed and how to reason about it.

C5: Selecting an appropriate process mining tool. Another significant challenge that related to the design phase is the selection of appropriate tools and platforms. A wide range of open-source and commercial tools, such as ProM, Disco, Apromore, Celonis, among others, is available. Each tool is similar in terms of their high-level purpose, which is to discover and analyze work processes; still, they are different in terms of features, usability, learning curve, and technical complexity. Participants highlighted that, particularly in academic settings, there is growing awareness that students should gain familiarity with commercial platforms, as organizations are more likely to adopt these tools in operational settings rather than using open-source libraries such as PM4Py.

However, this shift toward commercial tools introduces several pedagogical challenges. First, some platforms offer extensive functionality, which can be complex and overwhelming, especially for novice learners, potentially shifting the course focus toward tool training rather than conceptual understanding. Second, many commercial platforms implement their own built-in algorithms for core tasks such as process discovery. This makes it difficult for educators to explain differences between algorithms output, especially when the algorithmic details are not transparent or configurable. As a result, teachers often need to supplement commercial tools with open-source alternatives, e.g., ProM or PM4Py, to provide a comprehensive and methodologically grounded learning experience.

4.3 Implementation and Evaluation of Courses

Next set of challenges emerges during the implementation and evaluation of educational courses. This phase includes the implementation, i.e., the actual delivery of education, including lectures and trainings, facilitation of practical exercises, interaction with students and learners; and, evaluation, which relates to assessing learner performance, ensuring alignment with objectives.

C6: Balancing between applicability and complexity. P5 emphasized that process mining is often oversimplified in teaching to prioritize ease of under-

standing and immediate applicability. However, real-life process mining projects involve complexity, including hidden challenges and assumptions that are frequently overlooked. This discrepancy between simplified instruction and practical reality can hinder learners' ability to effectively apply process mining techniques in complex, real-world scenarios.

Conversely, excessive emphasis on technical details and theoretical complexity can prolong the learning process, increase cognitive load, and potentially discourage learners, particularly those without strong technical background. Participant P3 highlighted that, especially in industry, lengthy courses often face challenges in maintaining learner engagement and motivation. This emphasizes the importance of carefully balancing rigorous content with engaging delivery to support effective learning and ongoing learner interest.

C7: Teaching the temporal aspect of the data. Educators must find ways to illustrate the temporal aspect of the data during the data preprocessing, event log quality assessment and interpretation of process performance metrics. Typically it is done through practical exercises and visualizations that make the dynamics of process execution more tangible for learners. However, several participants noted that adequately conveying the importance and complexity of temporal data remains a challenge. Misunderstandings in this area can lead to incorrect interpretations of process behavior, ultimately limiting learners' ability to leverage process mining effectively. P1 highlighted this challenge in the context of more advanced, predictive process mining techniques. As P1 shared, learners often struggle to formulate meaningful questions that consider the progression and timing of events. As a result, educators must not only teach how to read and process timestamped data, but also foster the ability to reason about how time influences process behavior and outcomes.

C8: Lacking potential value and use case examples for process mining. Participants from academia in our study noted that students often lack familiarity with the types of questions practitioners pose and how process mining is applied to address concrete business problems. This challenge is further reinforced by differences in tooling. Industry practitioners frequently use commercial tools with integrated dashboards, process connectors, and predictive capabilities. These tools support not only process discovery but also monitoring, decision support, and business transformation initiatives. As a result, students may miss out on understanding the full range of use cases and the practical reasoning that underpins professional process mining work.

C9: Validating process mining results. This challenge was highlighted by both academics and practitioners. In both, the validation of process mining results goes beyond technical metrics such as fitness or precision; it requires a broader interpretive effort that considers organizational context, stakeholder feedback, and the limitations of the data itself. Industry educators reported that it is often not feasible to dedicate sufficient time to iterative validation or to explore how changes in data preprocessing affect outcomes. One industry participant emphasized that a key difficulty lies in demonstrating to learners that small changes in how the event log is created, such as the selection of case identifiers,

activity granularity, or timestamp resolution, can significantly affect the analysis results and the types of insights that can be derived. While academic settings may allow for more in-depth exploration and iterative refinement of logs and models, corporate training programs rarely have the flexibility to explore multiple validation cycles or incorporate extensive feedback from domain experts.

5 Discussion

Based on our findings, the nine high-level challenges in teaching process mining, in this section we outline three key future research directions.

RD1: Develop process mining-oriented educational tools and environments for teaching. Currently, process mining tools, data, and practice materials are often fragmented and not purpose-built for learning contexts (challenges C3-5, C7, C9). Future research should focus on identifying requirements and specificities to support developing sandbox environments or interactive tools that support exploratory learning. It may include visualizing the effects of incorrect assumptions, improper configurations, and algorithmic choices, helping learners understand both theoretical and practical dimensions.

RD2: Develop adaptive and modular process mining curriculum models for diverse learner profiles. Process mining learners come from diverse academic and professional backgrounds, leading to varied expectations, skills, and learning needs (challenges C1-2). Future studies should explore skills required for teaching and learning process mining, as well as develop curriculum models for process mining that support adaptive and modular instruction, enabling alignment with both technical and business-oriented learners. This could include the development of modular course structures, role-specific learning tracks, and content that bridges technical and organizational perspectives.

RD3: Bridge process mining education and real-world practice. There remains a gap between academic teaching and real-world application of process mining (challenges C6, C8). Future work should investigate strategies for integrating real-life complexity into educational settings through the use of realistic industry-sourced event logs and practice-driven case studies. These efforts should reflect the full lifecycle of process mining projects, such as data preparation and stakeholder communication, to enable learners to acquire practical competencies, which are essential for effectively implementing process mining in organizations.

6 Conclusion

In this work, we investigated what are the challenges in teaching process mining. Based on a series of focus groups with experienced process mining educators from both academia and industry, we identified nine high-level challenges in teaching process mining, such as the lack of reusable teaching materials and aligning goals with diverse learners profiles. With the identified challenges, we derived a set of research directions aimed at addressing key gaps in process mining education,

such as development of tailored educational tools and environments, adaptive curriculum models, and stronger integration between academic instruction and real-world practice. Our study does not come without limitations. It is based on a limited number of focus groups and participants, which may affect the generalizability of our findings. While our qualitative analysis involved thematic coding, which is inherently interpretive and prone to subjectivity, we mitigated this by collaborative review and discussion. Nonetheless, we believe our study offers valuable insights for future research and development in process mining education. As future work, we continue this study by follow-up interviews with educators to deepen understanding of specific challenges and investigate potential solutions.

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